

Exercise - Week 5: MATH1061/7861

Duration: 30 minutes

3 Questions - 10 marks

Topics: Proof by Contraposition, Rational Numbers, Divisibility and Modular Arithmetic

Student Name:

Student number: |_|_|_|_|_|_|_|_|

*Your student number should be 8 digits and **not** start with an s. e.g. 41234567*

Problems. Prove or disprove the following statements.

- (1) **(4 marks)** For all integers m and n , if $m^2 + n^2 + mn$ is even, then m and n are both even.

- (2) **(3 marks)** $\forall m, a, b \in \mathbb{Z}_{>0}$, $m \mid (ab)$ if and only if $m \mid a$ and $m \mid b$.

- (3) (**3 marks**) There exists an integer x such that $x^2 \equiv 3 \pmod{5}$.